



Beyond supervised learning

Dr Cian M Scannell, Assistant Professor

Department of Biomedical Engineering, Medical Image Analysis group

Learning strategies

Supervised

Learning from examples (=training data) that are labeled with their desired outputs. The goal is to learn general rules that maps inputs to outputs.

Unsupervised

Learning from examples without labels. The goals are:

- Learning the probability distribution that generated a dataset
- Finding structure in data
- Reducing dimensionality → feature learning

Learning strategies

**Supervised
methods**

*Weakly
Supervised*

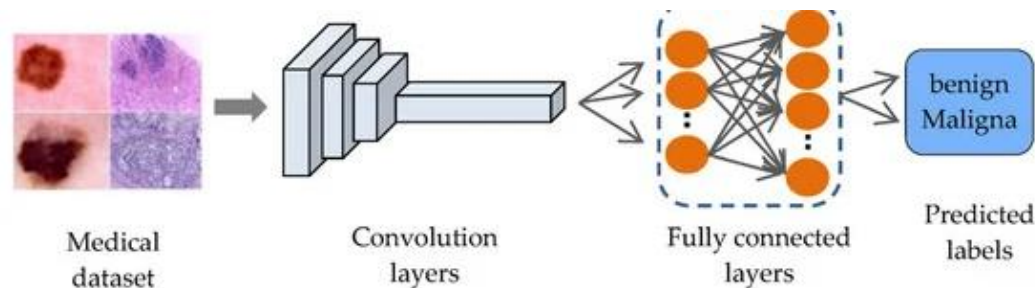
*Semi-
supervised*

**Unsupervised
methods**

*Self-
supervised*

Transfer learning

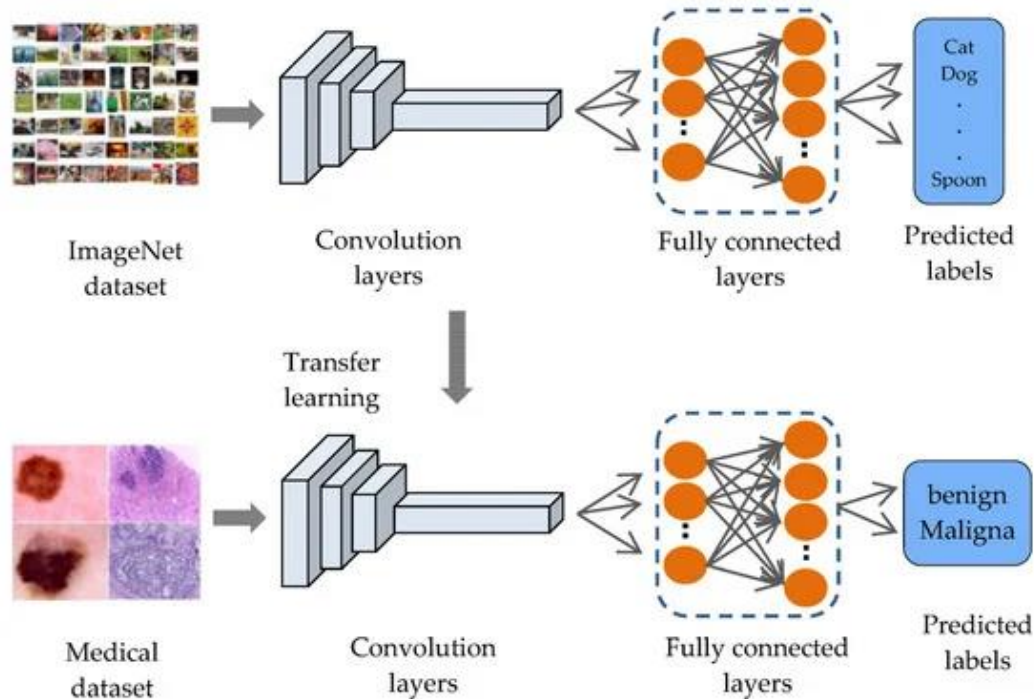
Motivation



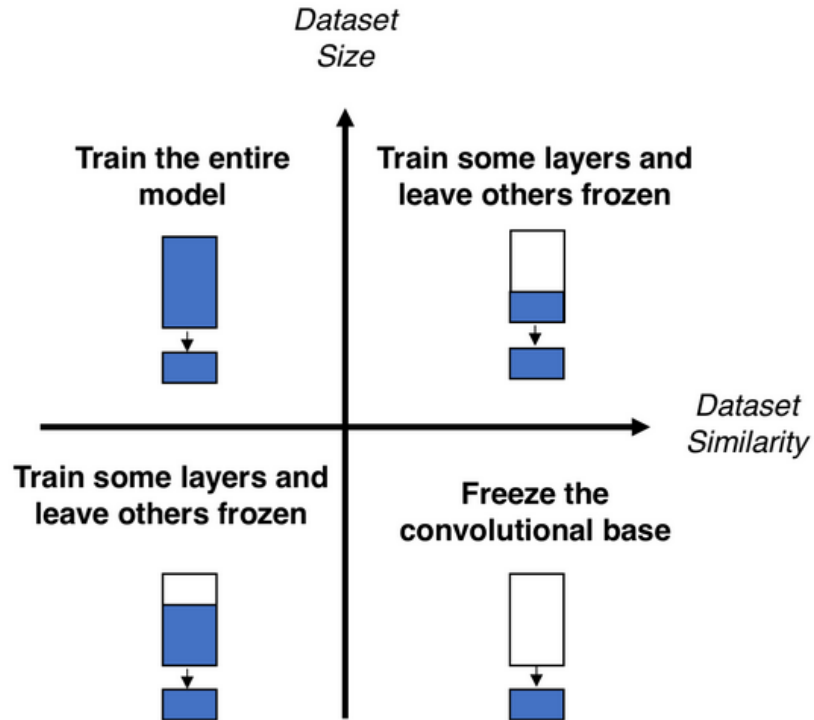
We want to train a model to classify skin lesions

But: These skin cancer types are rare and there is not much labelled data available for training

ImageNet– Transfer learning



Strategies



Formally

Given a source domain D_s with a corresponding learning task T_s and a target domain D_t with its corresponding learning task T_t .

Assuming that $D_s \neq D_t$ and $T_s \neq T_t$

Transfer learning is the process of improving the target task model $f_t(\cdot)$ by using the related information D_s and T_s

Strategies

Learn from simulations

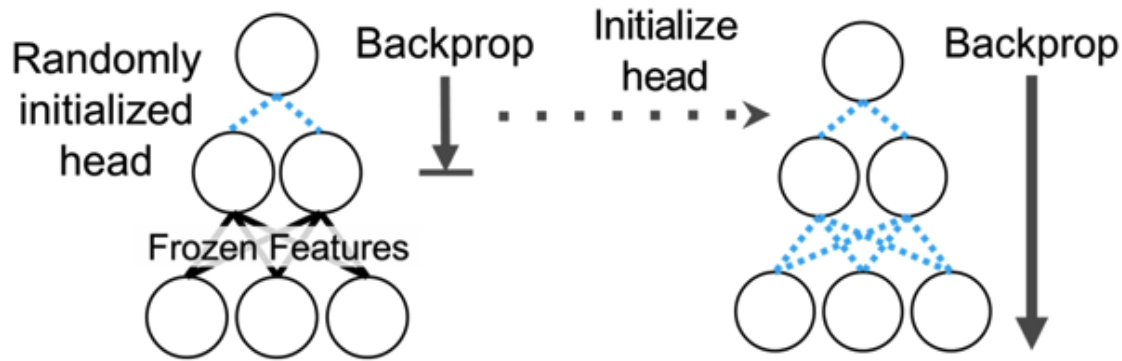


Domain shift

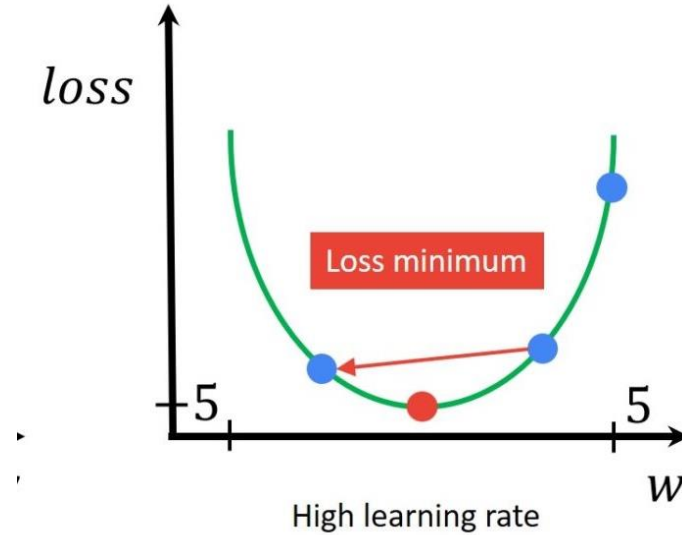


How to start

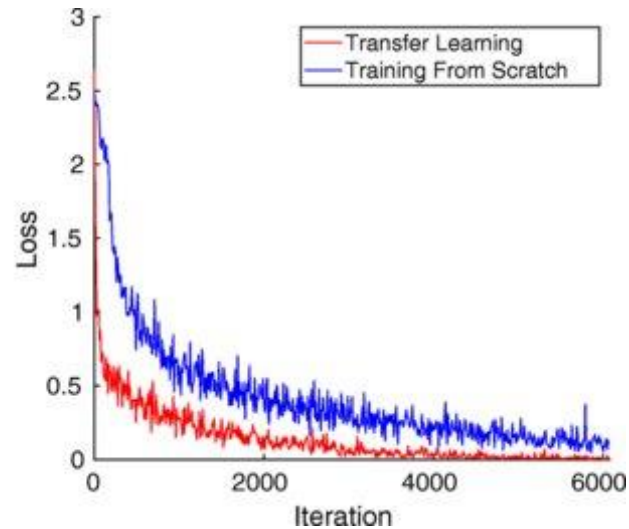
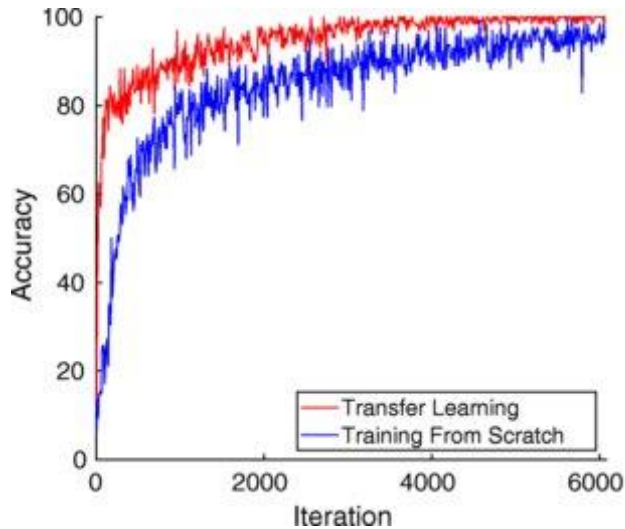
- Fix early layers and train last layer(s).
- Then fine-tune all layers



Lower learning rate



Transfer learning



Unsupervised learning

Applications

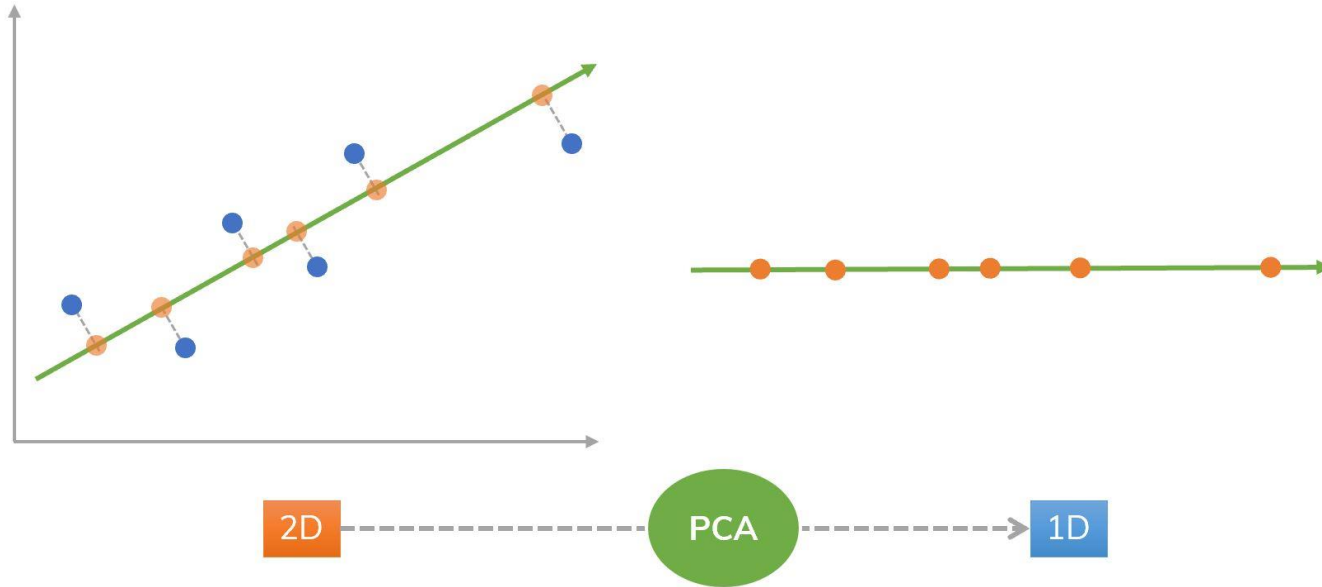
Dimensionality reduction

- PCA

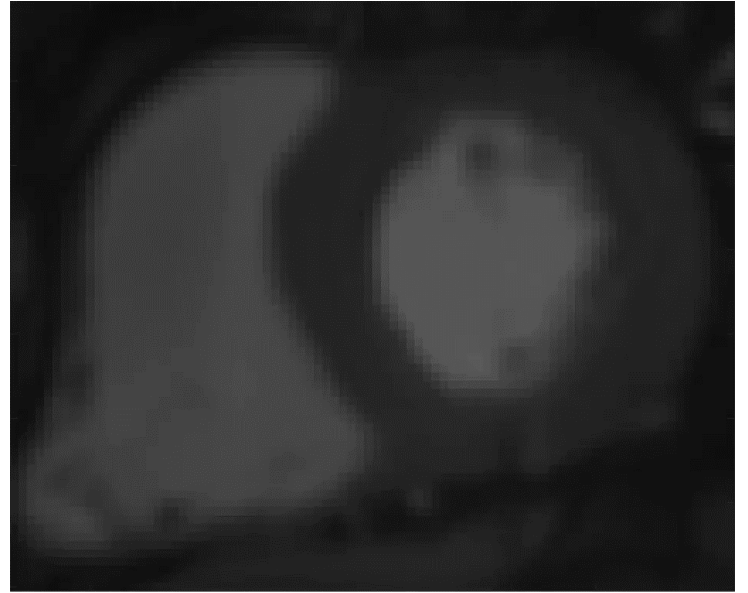
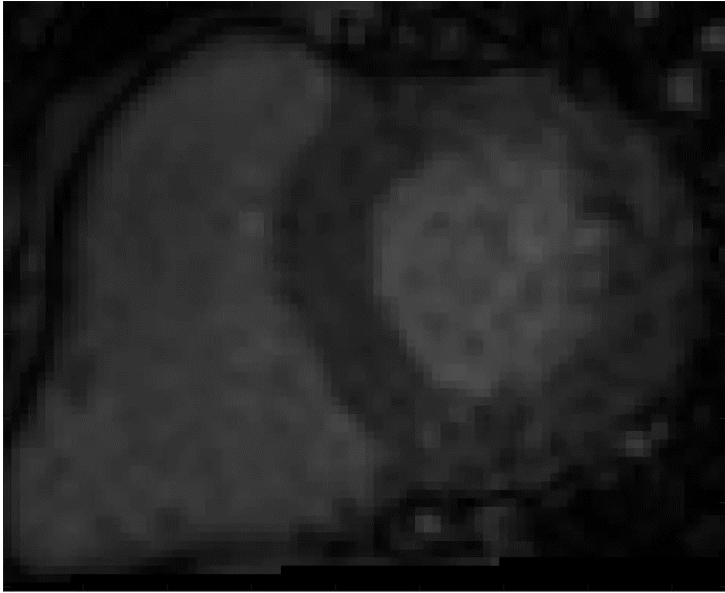
Clustering

- K-means clustering
- Hierarchical clustering

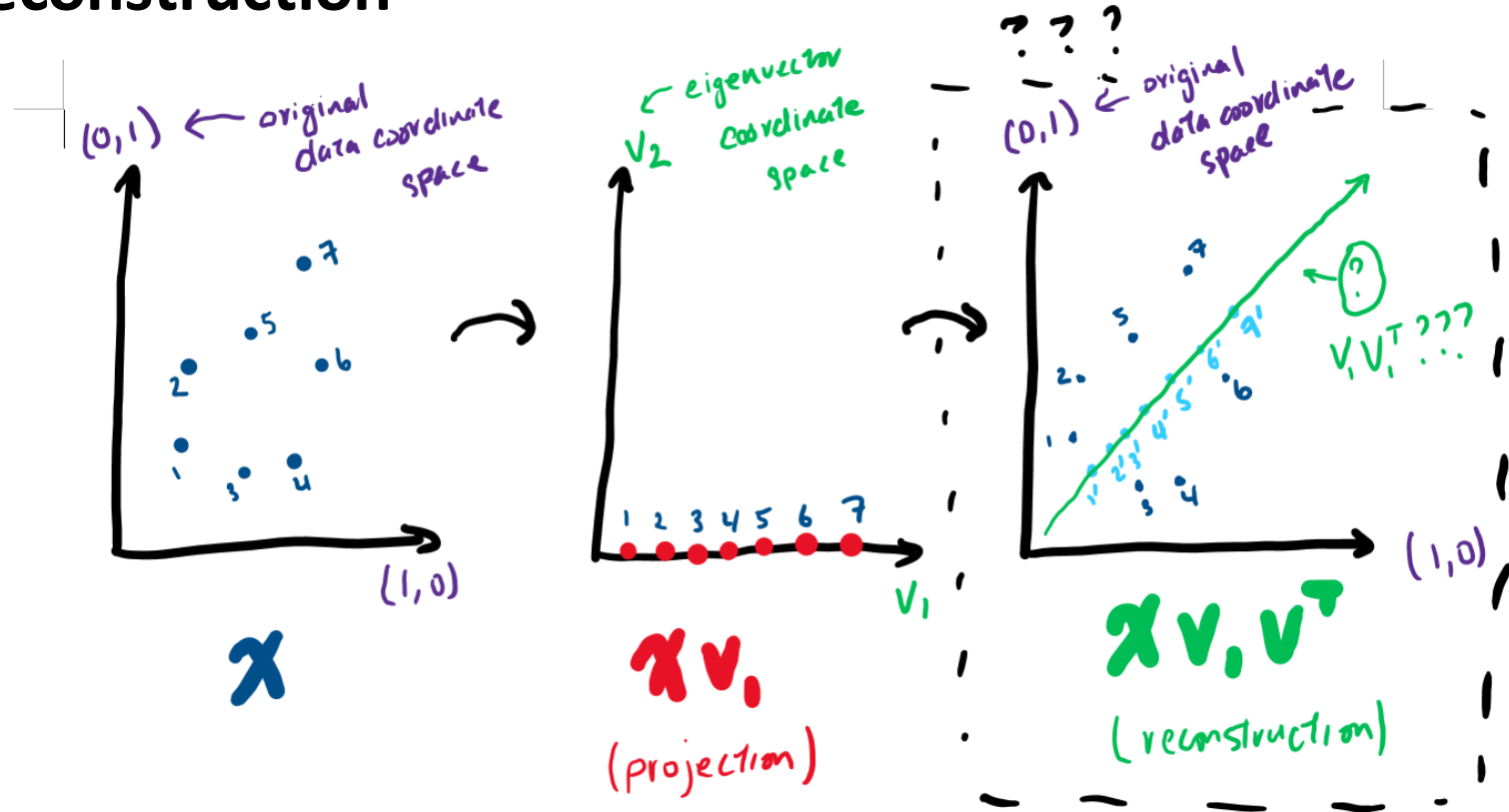
Dimensionality reduction



Application



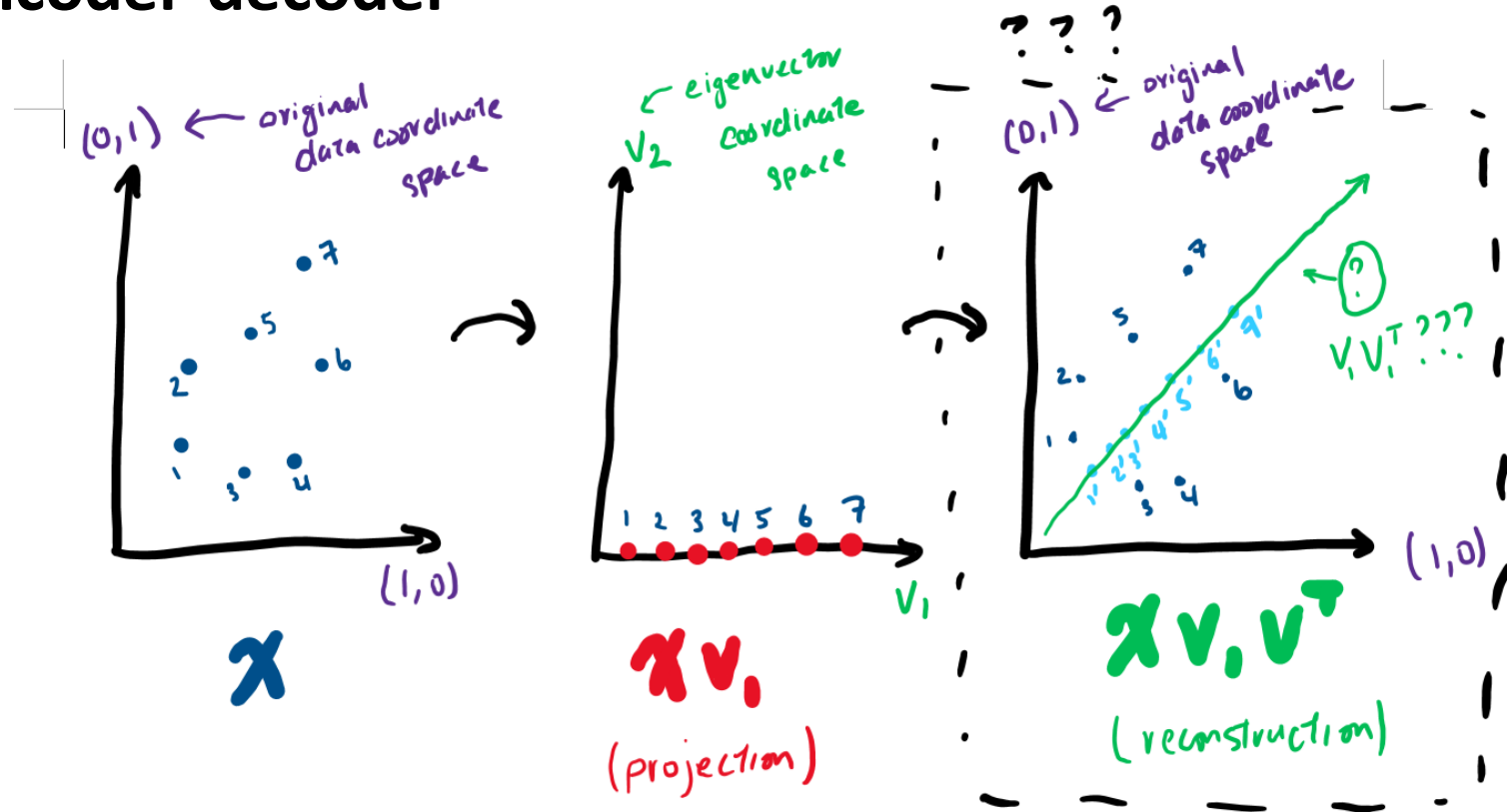
Reconstruction



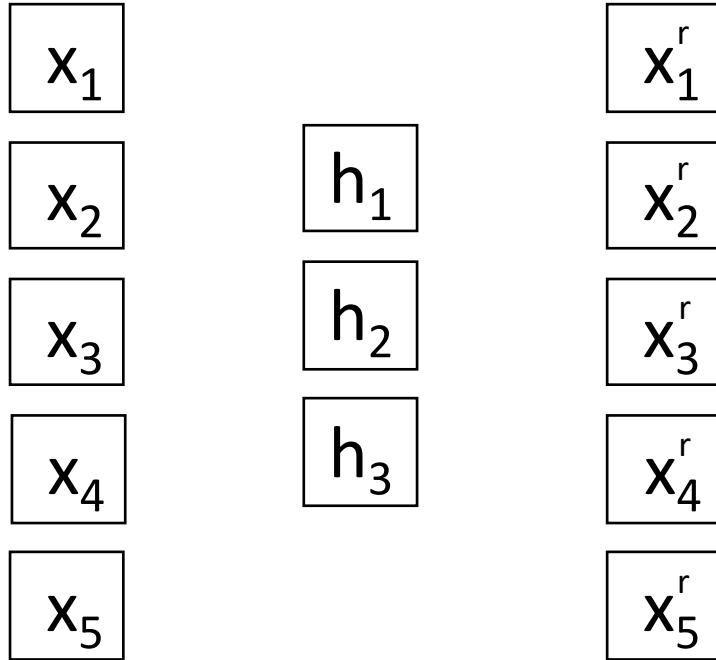
Limitations

- PCA assumes that the relationships between variables are linear. It may not perform well if the data has complex, non-linear relationships.
- The principal components are linear combinations of the original variables

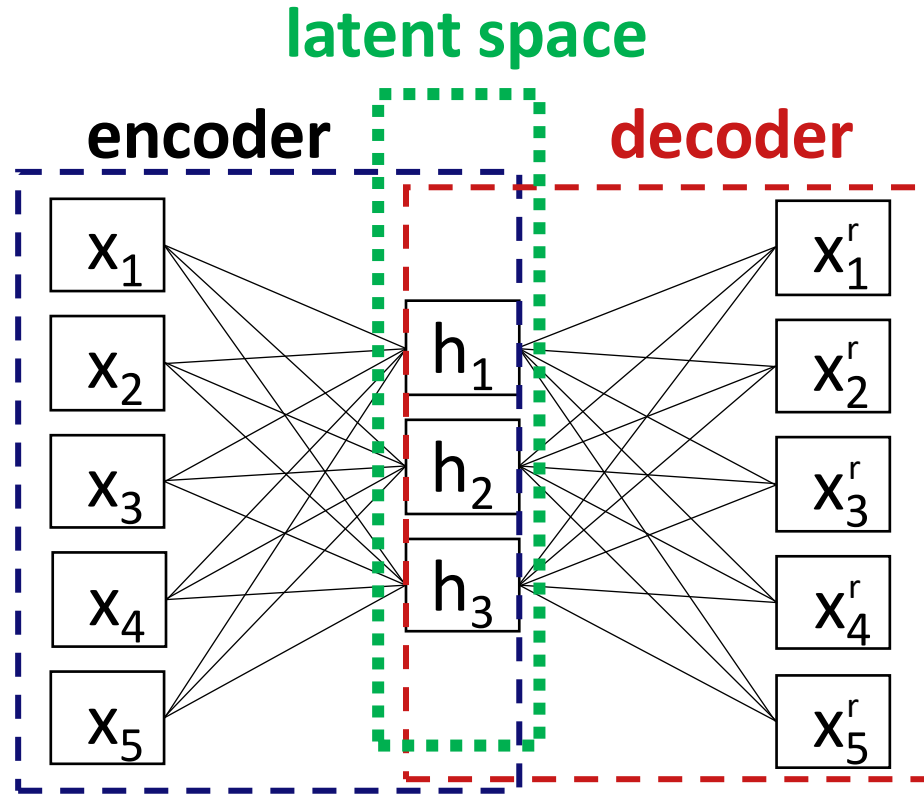
Encoder-decoder

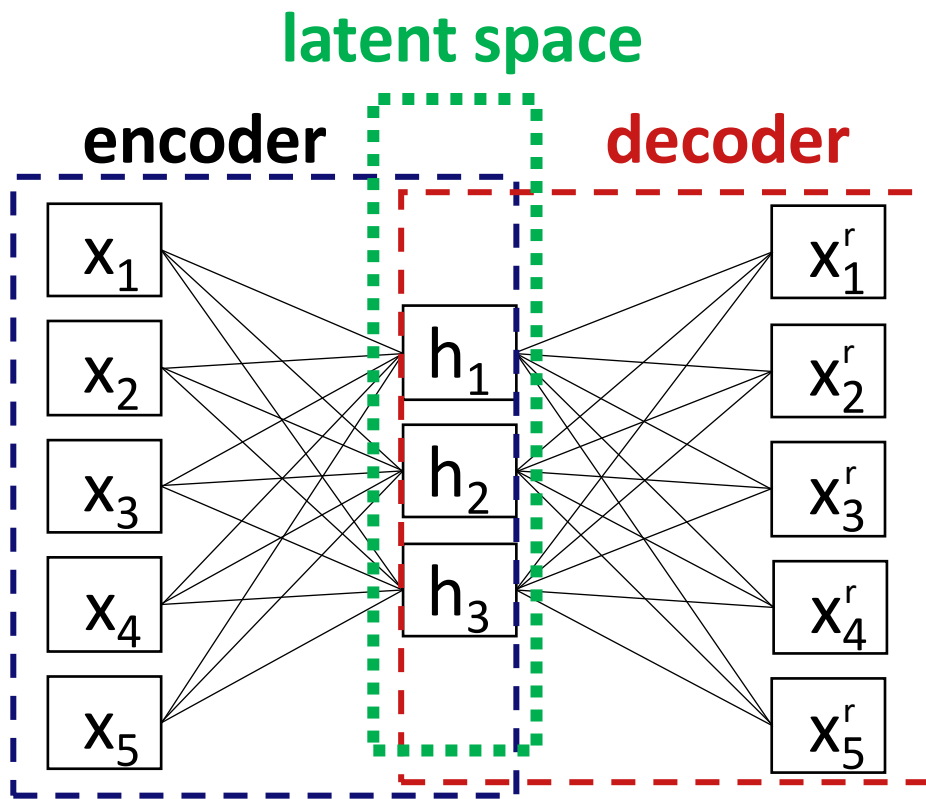


Autoencoder



Goal = reconstruct input x_i , using a restricted number of latent variables h_i





Encoder:

$$h = f(x)$$

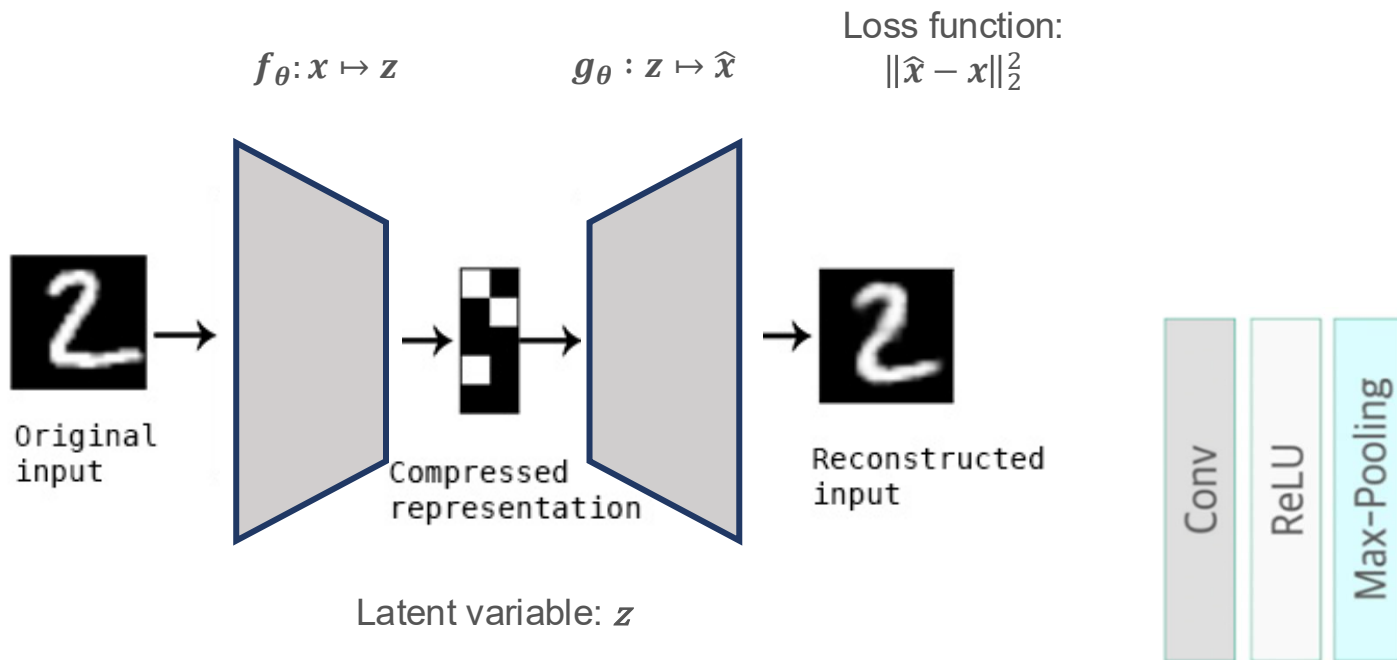
Decoder:

$$x^r = g(h)$$

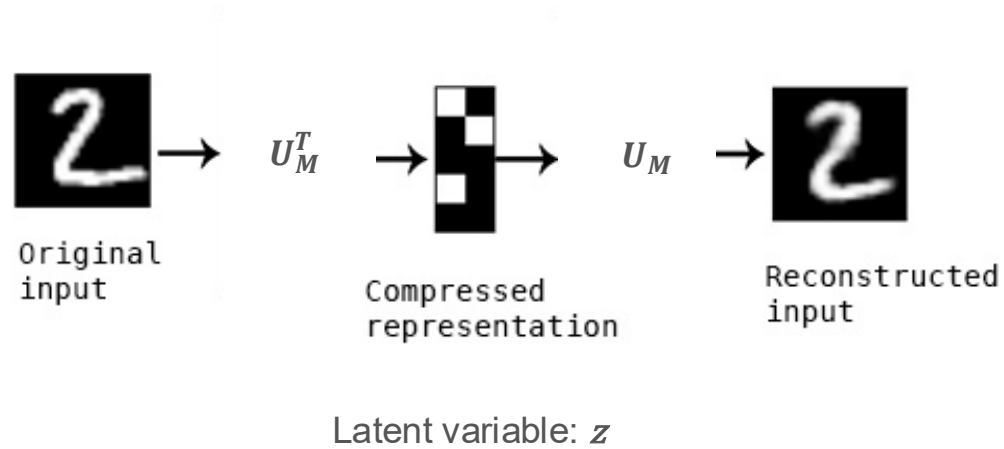
Penalize dissimilarity

$$L(x, g(f(x)))$$

Autoencoder (non-linear extension of PCA)



PCA



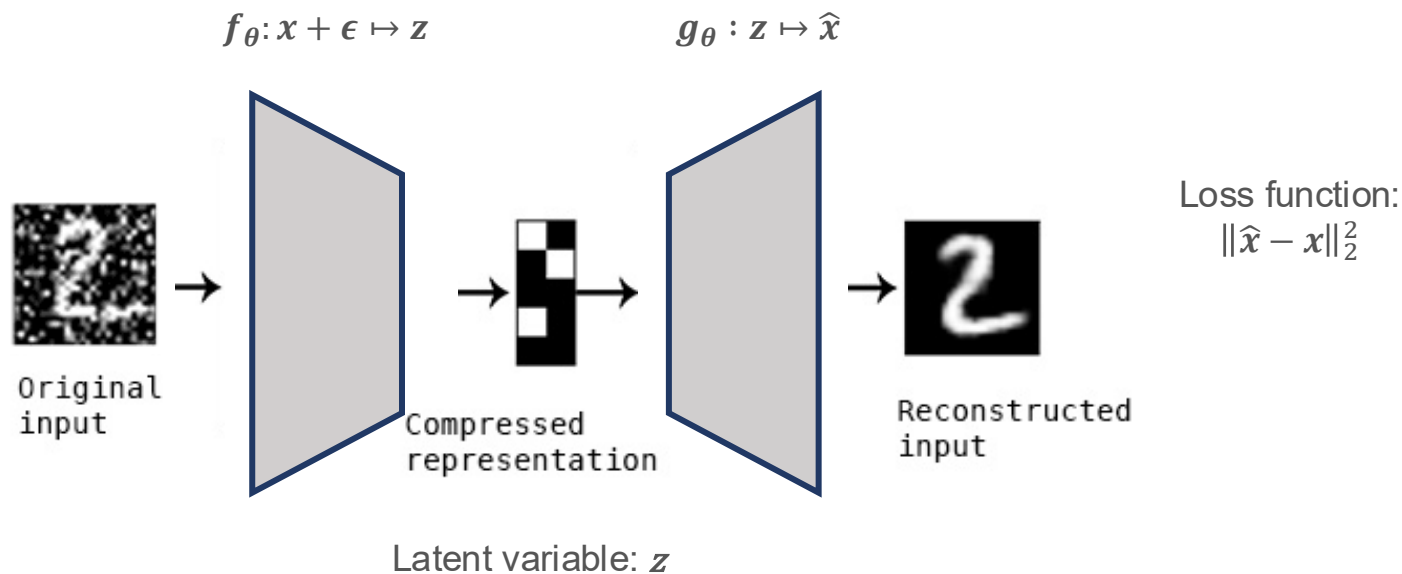
Autoencoder

- Encoder/decoder can be simple or complex
- For example: a deep convolutional neural network

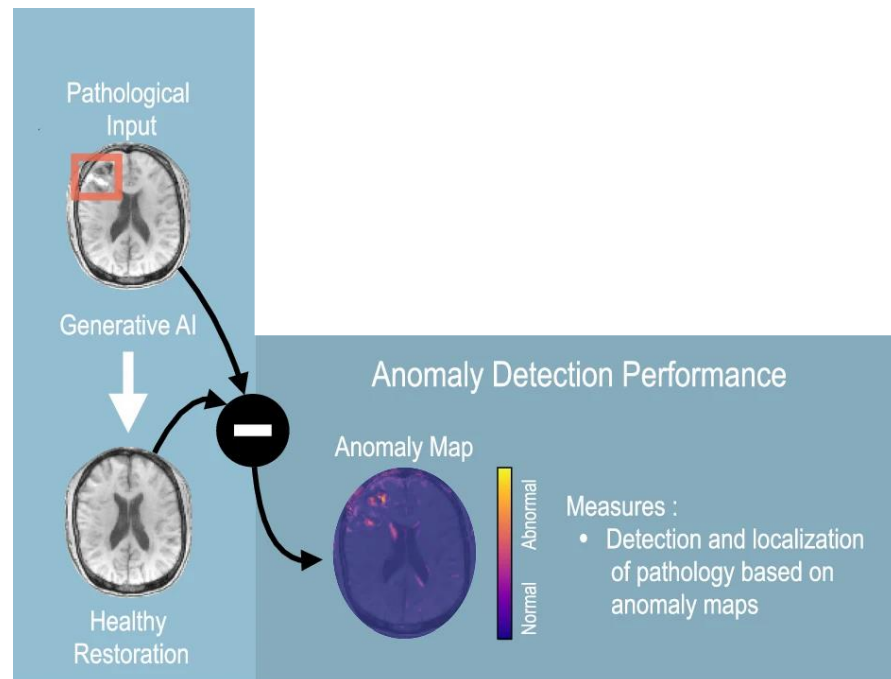
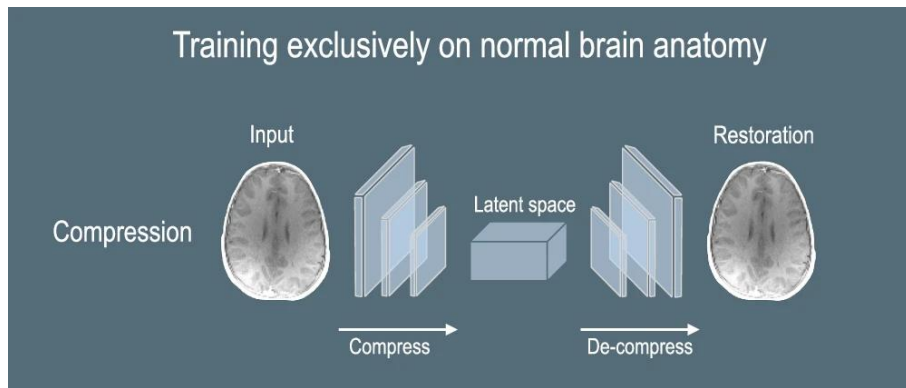
Applications

- Dimension reduction!
- Latent variables can be used for secondary objective, e.g. classification
- Denoising (by adding noise to the input and reconstructing the original)

Denoising Autoencoder

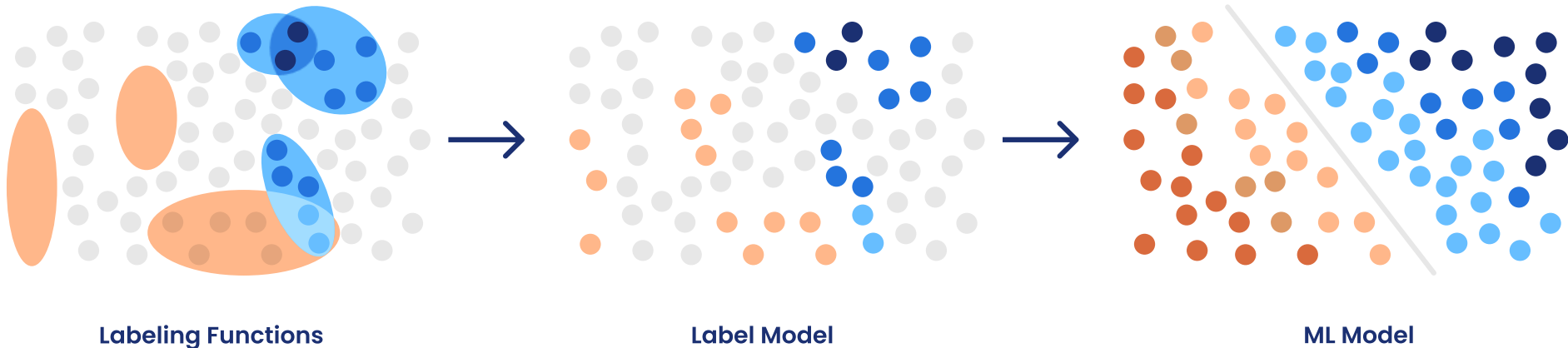


Normative representation learning

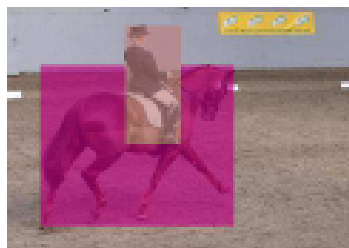


Weakly-supervised

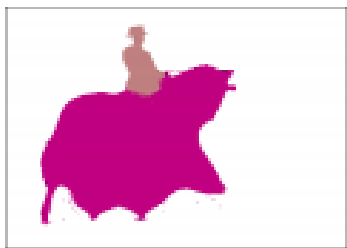
Weak-labelling



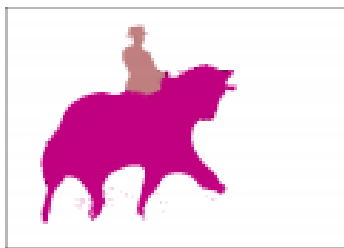
Bounding box annotation



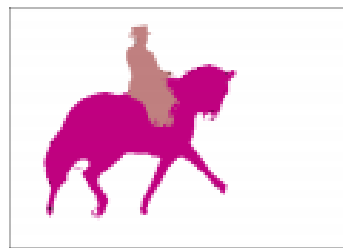
Example
input rectangles



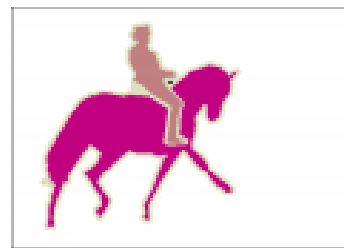
Output after
1 training round



After
5 rounds



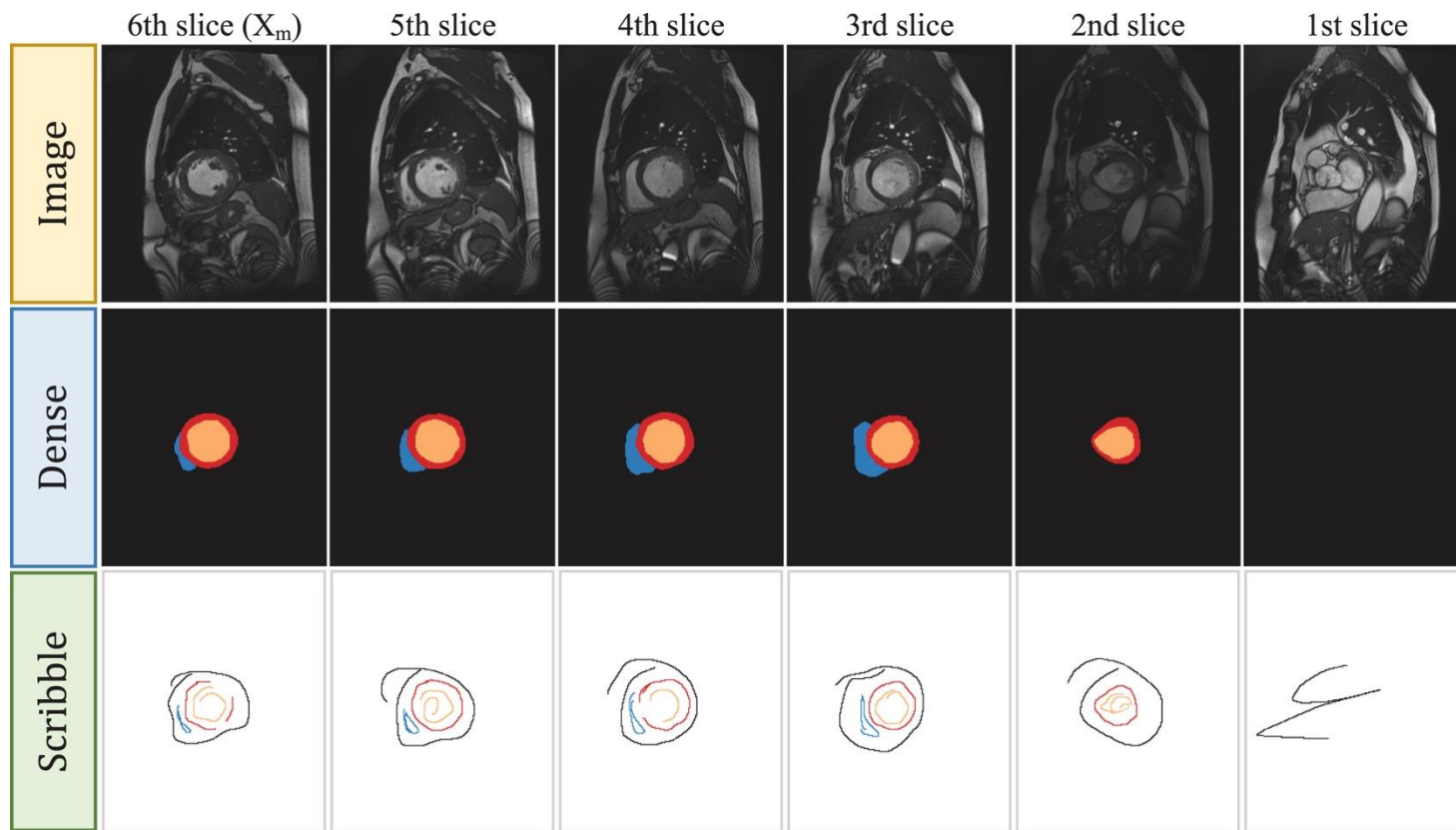
After
10 rounds



Ground
truth

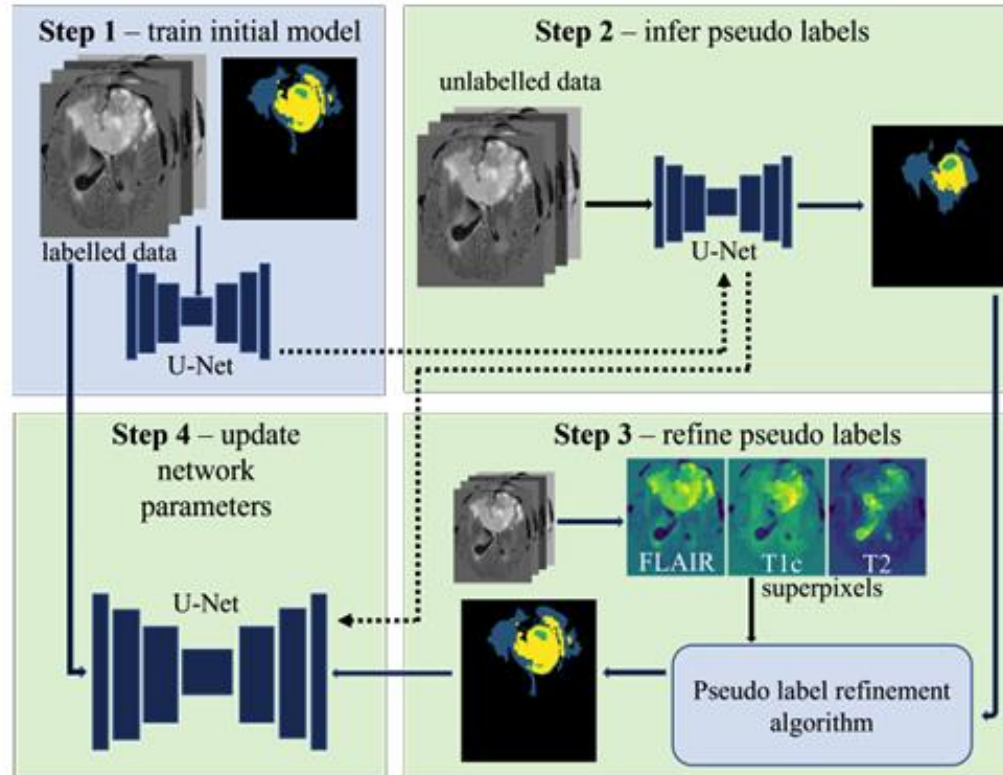
Khoreva et al. *Simple Does It: Weakly Supervised Instance and Semantic Segmentation*. CVPR 2017.

Scribble annotation



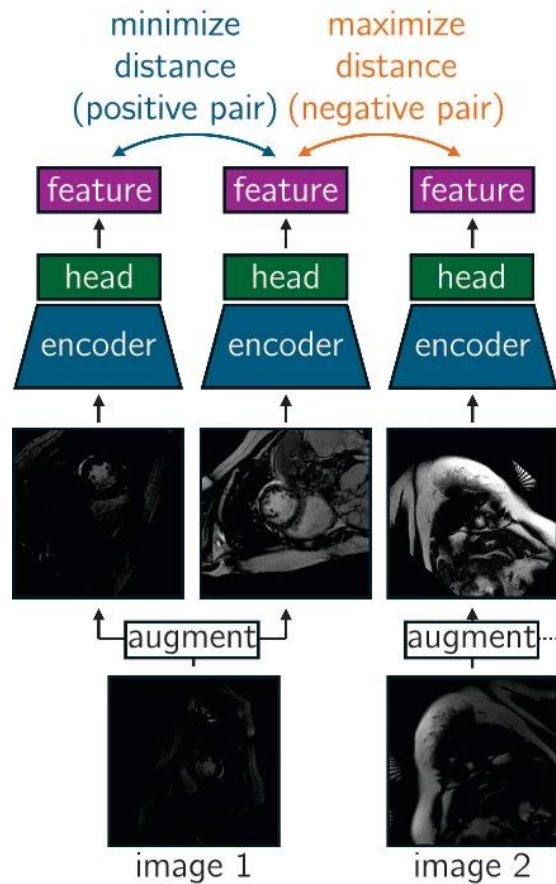
Semi-supervised

Pseudo-labelling



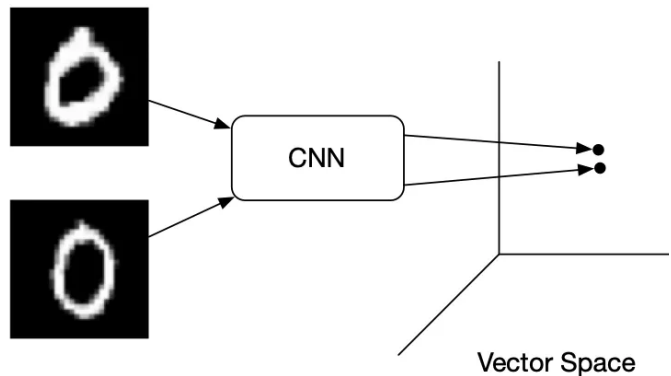
Self-supervised (pre-training)

SimCLR

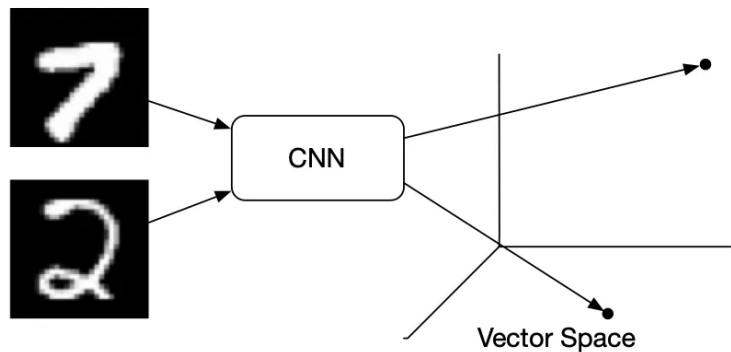


A Simple Framework for Contrastive Learning of Visual Representations

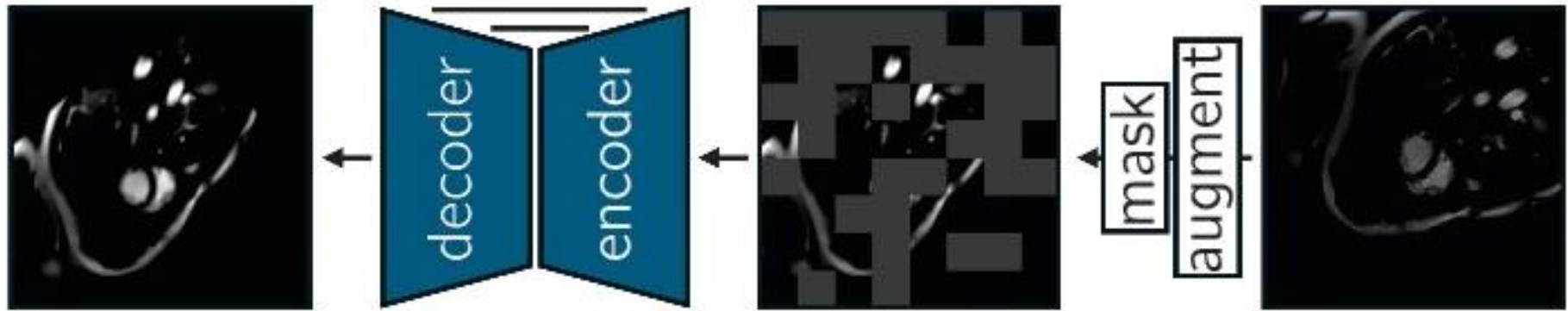
Contrastive loss



$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$$



Masked image modelling



Some remarks on self/semi-supervised learning

- Fewer labeled data needed
- For many medical applications data is still limited, e.g. because disease is rare
- Use knowledge from a related task to improve downstream task
- Humans also learn in a semi-supervised fashion

Thank you

c.m.scannell@tue.nl